

Analyzing the Impact of Border Conditions on Adversarial Patch Attack Performance

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Abstract. The vulnerability of object detection models to adversarial attacks has led to extensive research. Adversarial patch attacks are a key method for distorting model predictions. Various factors can affect the success rate of such attacks; consequently, existing studies have primarily focused on patch generation, neglecting elements such as patch borders and interpolation methods. This study analyzed the effect of various border conditions on the attack success rates of adversarial patches on object detection models. Experiments were conducted using patches generated from multiple datasets, applying different border thicknesses, colors, and interpolation methods to evaluate performance. The impact of the patch sizes and application ratio on attack performance was also examined. We suggest ways to maximize the effectiveness of patches by performing attacks under diverse conditions. This study highlights the importance of considering these factors in adversarial patch research to enhance the robustness of object detection models.

Keywords: Adversarial Patch Attack · Object Detection · Adversarial Attack Strategy · Human Detection

1 Introduction

Adversarial patch attacks are techniques designed to distort the performance or cause malfunctions in deep learning models, particularly visual recognition models [2, 6]. These attacks involve the application of patches to digital images or real-world objects [16]. The patches are crafted to prevent the correct recognition of objects, exploiting the vulnerabilities of deep learning models to achieve successful attacks [23]. Such patches can lead to unreliable results and pose significant risks in critical applications such as autonomous vehicles or security systems. For instance, in the case of autonomous vehicles, an adversarial patch applied to a road sign can cause the vehicle to misinterpret the sign, potentially leading to accidents [3, 14, 29, 30].

Patches are typically generated as follows. First, the image of the target object to be attacked is analyzed to identify the optimal pattern that will cause the model to malfunction when the patch is applied [5, 27]. This pattern is adjusted at the pixel level to prevent the object from being accurately recognized. To ensure robustness across different environments, patches are optimized for multiple factors such as brightness, angle, and size [11]. This ensures that the

patch works effectively regardless of different lighting conditions, viewing angles, and the sizes and shapes of the objects to which it is applied. This optimization process is designed to make the patch consistently induce malfunctions in the maximum possible number of situations.

However, the process of applying these patches to images often lacks consistency. For example, when applying patches to digital images, black borders can sometimes appear around the patch [24, 26]. In real-world scenarios, inconsistent borders may be used or thick borders might be added around the patch to avoid covering it with hands. These inconsistencies in the borders can negatively impact the effectiveness of the patch in disrupting recognition. If the border between the patch and the background image is inconsistent, the efficiency of the patch decreases, reducing the effectiveness of the attack. Therefore, maintaining consistency in terms of the borders of patches crucial. In our study, we defined borders as the visual borders appearing around the patch, emphasizing that maintaining consistent borders is essential for maximizing the patch effectiveness.

Therefore, our objective was to analyze how the presence, color distribution, and thickness of borders, as well as interpolation methods, affect the success rate of attacks when patches are applied to background images. Moreover, we aimed to demonstrate that the application process and borders are critical factors in the effectiveness of the attack.

We conducted experiments on patches that perform hiding attacks, which prevent object detection models from detecting an object, and altering attacks, which cause one object to be recognized as another. During the experiments, we examined how the thickness and color of the patch borders affected the performance of the attacks. For instance, we analyzed how the results varied with thicker borders or specific colors. Furthermore, we applied different interpolation methods that can be used when applying patches and investigated how each interpolation method impacts the effectiveness of the patch. Through these experiments, we comprehensively evaluated the effect of patch borders and interpolation methods on attack success rates (ASRs). Finally, we also examined how the changes according to the application ratio affected the outcomes.

In summary, the primary contributions of this work can be outlined as follows:

- We highlight the lack of consideration for patch borders in existing adversarial patch research and emphasize the importance of maintaining border consistency.
- We suggest ways to maximize the attack performance of patches by performing patch attacks under different conditions, such as varying the patch border thickness and color and using different interpolation methods, and analyzing the impact of these factors on the ASR.
- We analyzed how the border of a patch affects the ASR depending on the patch size or patch attachment ratio.

2 Background

2.1 Adversarial Patch Attack

Adversarial patches are small image patterns that are designed to deceive machine learning models, particularly image classifiers or object detectors. These patches are created using various loss functions and optimization techniques to disrupt the predictions of the model.

The initial image for the patch is typically obtained by applying random noise or a neutral color (e.g., gray). To update and optimize the patch, various loss functions are defined. The main loss functions include: classification loss, which minimizes the difference from the target class, objectness score loss [24], which reduces the probability of detecting objects in object detection models, total variation (TV) loss [19], which maintains the visual naturalness of the patch, non-printability score (NPS) loss [22], which ensures that the patch appears natural when printed or displayed, and saliency loss [5], which encourages the patch to be placed in less important areas of the image.

The patch is then applied to the image with various transformations such as different positions, sizes, angles, and brightness values to enhance its robustness. The patched image is then fed into the model to obtain prediction results, and the values of the defined loss functions are calculated. The patch is updated to minimize the loss using gradient descent. This process is repeated until the patch is optimized. The final patch is evaluated on various images and scenarios to assess how effectively it deceives the target model. Through these steps, the generated adversarial patch is optimized to effectively deceive the target model in diverse environments.

Thus, loss functions and transformations applied during patch creation are primarily considered in studies on adversarial patch attacks. However, this paper focuses on differences in interpolation methods and border conditions that can occur when applying the patch.

2.2 Patch Attack for Vision Task

Classificaion. Brown *et al.* [2] introduced adversarial patches, which are small patches that are applied to images to deceive deep learning models without needing access to training data. Their study indicated that attacks can be effective even without knowledge of the target model. However, these patches must be relatively large and noticeable, and their effectiveness can decrease under varying real-world conditions. Karmon *et al.* [10] proposed LaVAN, which uses localized and visible adversarial noise instead of traditional global, imperceptible noise. In this method, less than 2% of image pixels are covered without obscuring the main objects, achieving high success rates. Despite its effectiveness, this approach is highly image- and location-dependent, and small shifts can revert the classification. Liu *et al.* [15] proposed perceptual-sensitive generative adversarial network, utilizing generative adversarial networks to create more natural and less noticeable adversarial patches. While effective, this model struggles with

consistent performance because of its dependency on specific images and locations. Zhou *et al.* [31] introduced DiAP, thus performing attacks without training data knowledge by manipulating intermediate model activations and converting non-targeted attacks into targeted ones.

Object Detection. DPatch, proposed by Liu *et al.* [16], can attack both bounding box regression and object classification in object detection tasks. Lee and Kolter [12] proposed a patch that is robust to real-world lighting and location changes. Madry *et al.* [18] introduced projected gradient descend, ensuring attack effectiveness by maintaining the pixel values within the RGB range. Song *et al.* [23] proposed the first physical-world patch for object detection, introducing attacks to hide objects or create fake detections. Thys *et al.* [24] developed patches that were effective against various human shapes and colors to fool surveillance cameras. Hollander *et al.* [8] proposed camouflage patches to hide military assets from aerial detection. Lu *et al.* [17] developed Patch-Noobj, which can be resized to fit aircrafts. Wang *et al.* [25] proposed invisibility patches to hide specific objects from detectors.

Each study proposes and demonstrates effective attack methods under specific scenarios and conditions, showing their applicability in both the physical world and digital environments. However, the various situations that can arise when applying patches have yet not been researched. Therefore, our paper analyzes the often-overlooked aspect of borders when applying patches and, based on this, provides guidelines for increasing the ASR of adversarial patches.

3 Approach

3.1 Motivation

Research on patches has primarily focused on reducing their size or making them less noticeable. While prior studies have aimed to increase ASRs by adjusting the characteristics of the patches, they have not sufficiently considered the pixels and borders between the background image and patch. Existing studies show inconsistencies, with some considering borders and others not, and the application of borders when applying physical patches. Without consistently considering various border conditions, ASRs and reproducibility can be negatively affected. Therefore, maintaining border consistency is crucial to maximizing ASRs and achieving reproducible attacks.

Thys *et al.* [24] proposed an attack that can significantly reduce the accuracy of human detection models by generating adversarial patches that target different shapes of people, and showed that it works effectively even when captured on camera in real-world situations. However, patches that are applied in both digital and real-world environments have noticeable white borders. Adversarial patches update individual pixels in detail as they are generated, and the noticeable borders can have a negative impact on the ASR. Patches with inconsistent borders may actually facilitate detection rather than confuse the model.

Wang *et al.* [26] demonstrated that, unlike a fixed digital patch, a patch in a physical environment works effectively even when it is rotated or conditions change. However, in this study, the physical patch was displayed on the screen of an electronic device (liquid-crystal display), with a wide black border around the patch. This border can affect the ASR, as the presence of the border can enable the object detection model in detecting the patch more easily.

Xu *et al.* [28] proposed adversarial T-shirts using white background shirts to evade human detectors and withstand posture changes of moving people, modeling the effects of deformation on irregular objects such as T-shirts for the first time. Hu *et al.* [9] later suggested a method for achieving high attack performance with natural appearance by applying adversarial patches to black T-shirts. However, these studies did not consider the impact of background color on ASRs. Background color can affect patch visibility and reduce effectiveness. Therefore, patch performance must be evaluated across various background colors to develop more sophisticated and robust adversarial patches.

3.2 Analyzing Patch Borders

Based on these findings, we aimed to systematically analyze border conditions to identify important factors that affect the success of adversarial attacks. Specifically, we conducted attack experiments under different conditions, such as the type of interpolation, patch sizes and ratios, border thickness, and border colors to determine the optimal settings that can improve the performance of the attack in a digital environment. The conditions we chose are ones that can be set as hyper-parameters or added arbitrarily by an attacker during the process of applying an existing adversarial patch. In particular, there is no existing analytical research on these settings and they are not given much consideration by attackers. Therefore, our approach will highlight the importance of consistent border conditions in patch design and provide concrete guidance to develop effective adversarial patches.

Interpolation Methods. To ensure that an adversarial patch adheres to a fixed proportion within the bounding box of the object in the original image, re-sizing is necessary, which is achieved through interpolation methods [1, 7, 20, 21]. Interpolation alters each pixel value within the patch, affecting its visual quality and border smoothness, significantly impacting the ASR. We experimented with various interpolation methods. Because the border between the patch and original image varies with each method, we investigated how the choice of interpolation affects the integration and ASR.

Patch Sizes and Ratios. When generating adversarial patches, the patch size must be specified in pixel units. As the patch size increases, a greater variety of colors and shapes can be represented within the patch. Thus, the characteristics of the final patch can vary greatly depending on the initial size set during patch creation. Moreover, the attacker determine what proportion of the object the

patch will cover during the application process. Generally, the larger the proportion of the patch on the object, the higher the ASR. Therefore, we investigate how the attack success rate changes in each scenario with and without borders by setting up experimental environments with various patch sizes and ratios.

Border Thickness and Colors. We analyzed how the ASR varies as the thickness of the patch border changes. The experiment was conducted so that the maximum thickness of the border line was not excessive as compared to the width of the patch, ensuring that the border was less noticeable than the patch itself. Moreover, we set the border conditions as black, white, and gray, which are assumed to be close to the typical colors of the patch, to examine how the ASR of the patch changes with each color.

4 Experiment Design

In this paper, we investigate the impact of borders on the ASR by conducting an experiment to attack human objects in a digital environment by applying an adversarial patch on them. Thus, we can assume that a human is holding an adversarial patch.

4.1 Experimental Setup

Datasets. In this study, we used the Microsoft Common Objects in Context (COCO) dataset [13] and the INRIA Person dataset [4] to train and evaluate our patches. The COCO dataset comprises 80 object categories covering a variety of everyday scenes and contains approximately 118,000 training and 5,000 validation images. We selected images that include the “person” class from the COCO validation set for our experiments. In Tab. 1, COCO refers to these “person” class images. The INRIA dataset comprises human full-body data, consisting of 614 training and 288 test images.

In addition, We evaluated adversarial patch attacks using two You Only Look Once (YOLOv5) models: one trained on the COCO (person) dataset and the other trained on the INRIA dataset.

Patch Generation. The goal of the hiding attack was to prevent the YOLOv5 model from detecting humans. To achieve this, we generated and tested four types of patches: COCO_COCO, INRIA_COCO, INRIA_INRIA, and COCO_INRIA. The initial patch color was set to gray, and the loss functions comprised saliency loss, TV loss, NPS loss, and objectness score, as mentioned in Sec. 2.1. The loss weights were set to 1.0, 2.5, 0.1, and 3.0, respectively, to reduce the objectness score of the person class. Patch transformations were applied as referenced in Sec. 2.1. Examples of the INRIA_INRIA hiding patches that were generated with different application ratios are shown in Fig. 1.

Table 1: Four types of hiding patches that were used in the experiment. Hiding patches were generated to prevent the You Only Look Once (YOLOv5) model from detecting human objects by training it on the train datasets. The generated patches were applied to the attack datasets to perform the attack.

Object Detector	Patch Name	Train Datasets	Attack Datasets
YOLOv5	COCO_COCO	COCO	COCO
	INRIA_COCO	COCO	INRIA
	INRIA_INRIA	INRIA	INRIA
	COCO_INRIA	INRIA	COCO

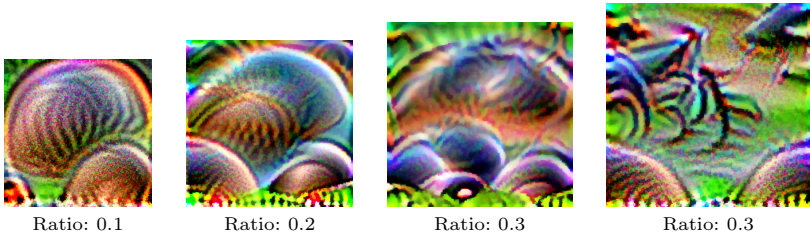


Fig. 1: Three examples of generated INRIA_INRIA hiding patches.

Altering attacks are classified into two types: targeted and untargeted. A targeted attack misleads the model to predict a specific target label, whereas an untargeted attack predicts one of several possible labels. We created untargeted patches that were predicted as various classes such as teddy bear, kite, and traffic light. The loss weights were set to 0.25, 0.5, 0.1, 8.0, and 6.0 for saliency loss, TV loss, NPS loss, original classification score, and target classification score, respectively, to reduce the classification confidence score for the person class and increase it for the target class. The target class was set to the one with the lowest classification score at each epoch. Other training conditions were the same as those for the hiding patches. Fig. 2 shows examples of untargeted altering attack patches that were predicted as various classes.



Fig. 2: Three examples of generated altering patches.

4.2 Border Conditions of Adversarial Patch

Interpolation. An important factor to consider when applying a patch is interpolation. Interpolation adjusts the image by calculating new pixel values to match the patch size to the object size in the original image [1, 7, 20, 21]. Various interpolation methods have been developed such as linear, cubic, and nearest neighbor, each producing different border colors for the patch. This is because each method processes the pixel values at the patch edges differently.

Note that a binary mask is used to ensure proper alignment of adversarial patches with objects in images. This mask marks the area of the object with white pixels (value 1) and the external area with black pixels (value 0). During application, the patch is placed only on the white pixel areas, ensuring precise placement and minimizing any mismatch in pixel size or position. The various interpolation methods are explained subsequently.

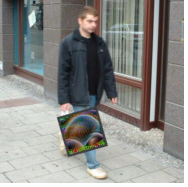
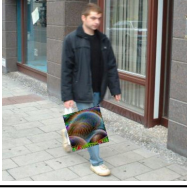
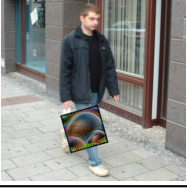
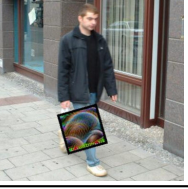
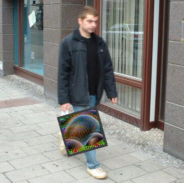
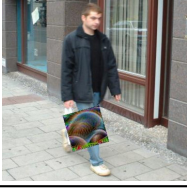
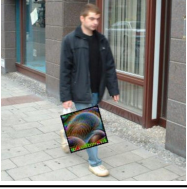
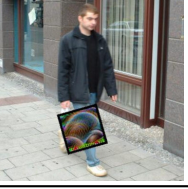
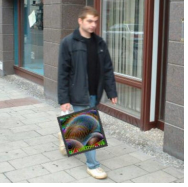
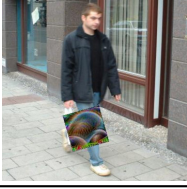
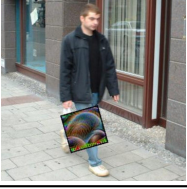
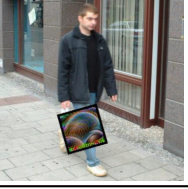
- **Bilinear method:** It uses the ratio of the pixel values and distances of the four neighboring pixels to calculate the unknown pixel value. As the binary mask of the adversarial patch sets the area outside the patch to zero, several zero-value pixels appear around the border, making it appear black.
- **Cubic or bicubic method:** It estimates the value of each new pixel by calculating a weighted average of its 16 nearest neighbors, with closer pixels having higher weights. Similar to bilinear interpolation, this method also results in black borders owing to the zero-value pixels in the binary mask.
- **Nearest neighbor method:** It uses the known value of the closest pixel without considering other pixels. This method fills the border with the closest pixel value, making the border less noticeable as compared to that in the case of the bilinear or bicubic interpolation.

Color of the Borders. To analyze the effect of border conditions on ASRs, the border colors were varied as white, gray, and black. Examples of hiding patches with different border colors in a patch size of 256×256 pixels are shown in Tab. 2, where the first, second, and third rows show the patches with white, gray, and black borders, respectively.

Thickness of the Borders. The border thickness was maintained below 6.25% of the patch dimensions to ensure that the borders did not excessively influence the visual perception of the patch itself. For patches of 256×256 pixels, the border thicknesses were set to 2 (0.78%), 8 (3.13%), and 16 (6.25%) pixels. The results of applying patches with different border thicknesses can be seen in each column of Tab. 2. Similarly, for patches of 128×128 pixels, the border thicknesses were set to 2 (1.56%), 4 (3.13%), and 8 (6.25%) pixels. For patches of 64×64 pixels, the border thicknesses were set to 2 (3.13%) and 4 (6.25%) pixels.

Various conditions that might occur when applying patches were considered and transformations were applied accordingly. Gaussian random noise was added, patches were rotated at varying angles, and they were applied to different locations on the object. Moreover, to simulate various scenarios, adjustments were made to the contrast and brightness values.

Table 2: Examples of patch applications according to color and thickness of the border.

		Thickness		
Color		2 pixels (0.78%)	8 pixels (3.13%)	16 pixels (6.25%)
White				
				
				
Gray				
				
				
Black				
				
				

4.3 Evaluation Metric

The hiding patch attack was considered to be successful when objects labeled as “person” were misclassified as “background.” The hiding ASR is the proportion of detected “person” objects that were misclassified as “background” out of the total detected “person” objects in the dataset. The following equation was used to determine the hiding ASR:

$$ASR_{hiding} = \frac{N_{person \rightarrow background}}{N_{person}}, \tag{1}$$

where $N_{person \rightarrow background}$ represents the number of “person” objects that are misclassified as “background” and N_{person} denotes the total number of detected “person” objects.

The altering patch attack was considered to be successful when objects labeled as “person” were misclassified as a targeted altering class. The altering ASR is the proportion of detected “person” objects that were misclassified as the target altering class out of the total detected “person” objects in the dataset. To compute the altering ASR, we used the following formula:

$$ASR_{altering} = \frac{N_{person \rightarrow target}}{N_{person}}, \tag{2}$$

Table 3: Comparison of ASRs by patch types and interpolation methods. The used patches are 256×256 in size hiding patches.

Ratio	0.1		0.2		0.3		0.1-0.3	
Interpolation	Bicubic Nearest		Bicubic Nearest		Bicubic Nearest		Bicubic Nearest	
COCO_COCO	16.8	16.8	39.7	38.2	63.9	62.9	60.4	58.8
COCO_INRIA	45.2	44.8	66.9	64.7	92.8	91.6	92.3	91.3
INRIA_COCO	7.4	7.3	32.5	31.2	72.9	71.2	55.0	54.1
INRIA_INRIA	22.1	22.1	77.7	75.3	96.2	95.8	94.7	94.2

where $N_{person \rightarrow target}$ represents the number of “person” objects that are misclassified as the target altering class and N_{person} denotes the total number of detected “person” objects.

5 Results

5.1 Impact of Interpolation Methods

When applying the adversarial patch to the object, the bicubic method applied the patch with a black border around it, whereas the nearest neighbor method applied the patch without a black border. In the binary mask of the adversarial patch, the area outside the patch was set to 0, resulting in several pixels around the border having a value of 0, resulting in a border with a black appearance. In the case of bilinear interpolation, the results were identical to those of bicubic interpolation. Tab. 3 lists the results of the different interpolation methods for different patches on the training and attack datasets. The patches used here had sizes of 256×256 pixels. We used the aforementioned four types of hiding patches and varied the percentage of patches that were applied to objects.

The results indicate that, regardless of the patch type, the ASR is higher when using bicubic interpolation. The difference was up to 2.4%, with the ASR ranging from 75.3% to 77.7% when INRIA_INRIA patch was applied with a ratio of 0.2, suggesting that the black border enhances the effectiveness of the patch in the detection model. This implies that even small changes in pixel values due to interpolation can significantly impact the ASR. Therefore, to maximize the effectiveness of an adversarial patch, adding a thin black border around the patch can be beneficial.

5.2 Impact of Patch Sizes and Ratios

Tab. 4 and Tab. 5 list the results of the hiding and altering patch attacks under various border conditions, respectively. We compared cases with and without borders; for cases with borders, we examined the differences based on patch size and patch ratio. In these tables, bold texts indicate the maximum increase

in ASR compared to the no-border condition, while underlining indicate the minimum decrease in ASR compared to the no-border condition, or similar.

The ASR difference for hiding patches did not depend significantly on patch size. We defined “ASR difference” as the absolute value of the maximum difference between the case with no border and the cases with various borders. For example, when applying the INRIA_INRIA patch at a rate of 0.1, the ASR was 22.1% with no border on a patch of 256×256 pixels and 15.3% for a patch with a 16-pixel white border, exhibiting a difference of 6.8%. Similarly, for a 128×128 patch, the ASRs were 21.7% and 17.2% for the patches with no border and 8-pixel white border, that is, a difference of 4.5%. For a 64×64 patch, the success rate was 22.6% without a border and 15.3% with a 4-pixel white border, a difference of 7.3%. Furthermore, the 0.2 and 0.3 ratios also show no trends with respect to patch size, suggesting that the differences in the ASRs with and without borders is not affected by patch size. The ASR difference of the altering patch also did not vary significantly with patch size.

The ASR difference between the hiding patch with and without borders varies by up to 9.8%, with an average of 4.5%, depending on the patch application ratio. Considering that ASR difference is positive in most cases, except for a few special instances, the variation in ASR difference is primarily interpreted as an increase in ASR. For the six patches tested with an application ratio of 0.1, the INRIA_INRIA 64×64 patch showed the highest absolute value of the maximum difference at 7.3%, with an average value across all patches of 4.5%. Similarly, with an application ratio of 0.2, the maximum and average values are 9.8% and 5.1%, respectively, while at a ratio of 0.3, they are 7.2% and 3.4%. In addition, for altering patches, ASR difference varies by an average of 2.4% depending on the patch application ratio. Generally, when applying adversarial patches to objects, they are applied at a size between 0.1 and 0.3 of the object size, suggesting that patch borders can help improve the ASR.

5.3 Impact of Border Thickness

No consistent trend can be observed regarding the thickness of the borders. In hiding attacks, INRIA_COCO patches show higher ASRs with thicker borders, whereas INRIA_INRIA patches exhibit lower ASRs with thicker borders. In altering attacks, INRIA_COCO (kite) patches show no trend with respect to border thickness, whereas COCO_COCO (traffic light) patches exhibit higher ASRs with thicker borders. Therefore, a consistent relationship between border thickness and ASR could not be identified.

However, in general, using gray border with a thickness between 4 and 8 pixels clearly enhanced the ASRs. This specific thickness range and color combination consistently provided higher success rates under various conditions, making it a crucial strategic element to consider for improving ASRs.

Table 4: Hiding ASRs under various patch conditions.

Patch Type	Patch Size	Thickness	Patch Area Ratio								
			0.1			0.2			0.3		
			White	Gray	Black	White	Gray	Black	White	Gray	Black
INRIA_COCO	256 x 256	16	10.6	8.0	8.7	35.6	33.9	35.5	71.6	74.5	70.6
		8	10.0	7.9	8.3	35.5	34.0	<u>32.6</u>	70.8	72.8	<u>68.1</u>
		2	7.6	7.3	<u>7.0</u>	33.3	33.0	33.9	71.2	71.3	70.8
		None	7.3			31.2			71.2		
	128 x 128	8	8.5	<u>5.0</u>	5.9	32.1	29.4	30.3	66.9	66.4	<u>61.6</u>
		4	7.3	5.8	5.9	29.8	29.9	<u>28.0</u>	64.4	65.8	62.4
		2	6.9	6.9	7.3	30.2	30.5	28.7	63.9	63.7	62.1
		None	5.4			28.9			61.7		
	64 x 64	4	9.6	7.3	8.7	27.8	23.6	26.0	53.7	54.1	53.9
		2	8.3	<u>5.7</u>	6.8	24.6	<u>20.2</u>	23.3	50.0	<u>49.6</u>	51.0
		None	5.2			18.0			46.9		
INRIA_INRIA	256 x 256	16	<u>15.3</u>	21.9	18.5	<u>76.2</u>	80.2	76.8	<u>95.5</u>	96.2	96.2
		8	17.4	22.1	19.8	77.5	80.2	77.9	95.8	96.2	96.4
		2	20.0	21.5	21.7	78.9	79.2	78.1	96.2	96.2	96.4
		None	22.1			75.3			95.8		
	128 x 128	8	<u>17.2</u>	24.0	20.6	<u>68.5</u>	74.0	69.1	94.3	96.8	<u>94.2</u>
		4	18.9	22.5	22.3	68.7	74.3	71.5	94.9	97.0	96.0
		2	20.0	22.5	22.6	70.6	72.8	72.5	95.3	96.8	96.2
		None	21.7			71.7			96.4		
	64 x 64	4	<u>15.3</u>	22.1	20.6	60.0	64.2	<u>57.9</u>	<u>91.5</u>	92.8	92.1
		2	17.4	22.3	21.9	61.5	63.8	60.6	92.1	93.0	91.9
		None	22.6			63.0			91.1		

5.4 Impact of Border Colors

For the INRIA_COCO hiding patch, the ASR generally increased when the border color was white compared to that in the unbordered state, with an maximum increase of 9.8% and an average increase of 4.5%. Conversely, when the border color was black, the ASR was similar or reduced. For the INRIA_INRIA hiding patch, the ASR generally increased when the border color was gray, with an maximum increase of 4.9% and an average increase of 1.5%. In contrast, when the border color was white, ASR was similar or tended to decrease.

These results suggest that the characteristics of each patch are influenced by the dataset on which it was trained, reflecting the inherent properties of the patches. The INRIA_COCO hiding patch was trained with the COCO dataset, whereas the INRIA_INRIA hiding patch was trained with the INRIA dataset. As each patch was designed to be inconspicuous within its respective training dataset, we can infer that the color characteristics of the dataset affect the change in ASR based on the border color. This trend can also be observed in Tab. 5.

Table 5: Altering ASRs under various patch conditions.

Patch Type	Target Class	Patch Size	Thickness	Patch Area Ratio					
				0.2			0.3		
				White	Gray	Black	White	Gray	Black
INRIA_COCO Kite		512 x 512	32	4.3	4.8	<u>3.9</u>	11.5	12.2	<u>10.2</u>
			16	5.3	6.0	4.7	11.9	12.1	11.6
			8	4.3	5.7	4.4	12.2	12.1	11.8
			None	6.3			13.4		
COCO_COCO Traffic Light		128 x 128	8	9.3	7.9	8.1	20.5	20.4	20.0
			4	8.3	7.8	7.3	21.1	20.5	20.2
			2	7.8	7.2	<u>7.2</u>	20.6	20.0	<u>19.7</u>
			None	6.8			19.4		

All the altering patches were trained with the COCO dataset and, similar to the hiding patches, showed an increase in ASR with white borders, whereas the ASR remained the same or was reduced for the patches with black borders. Therefore, to maximize the effectiveness of the patch attack, border colors that reflect the color characteristics of the dataset must be used to train the patches.

Fig. 3 shows example images of the hiding attack results according to border color differences. The images are displayed in order of confidence, from highest to lowest (*i.e.*, from lowest to highest ASR). The patch used here is INRIA_INRIA, and the ASR is highest when the border is gray and tends to be lowest when the border is white.

The first row shows an example where a patch of size 128×128 at a ratio of 0.1 is applied to a person object with a border thickness of 8 pixels at a ratio of 0.1. The confidence score fall below 0.4 only when the border is gray, thus successfully performing a hiding attack; conversely, the confidence score when the border is white is 0.83, indicating the lowest attack performance among the failed attacks. The second row shows an example where a patch of size 128×128 at a ratio of 0.2 is applied to a person object with a border thickness of 8 pixels at a ratio of 0.2. In this case, the hiding attack is also successful only when the border is gray. Finally, the third row shows an example where a patch of size 128×128 at a ratio of 0.3 is applied to a person object with a border thickness of 2 pixels at a ratio of 0.3. In this case, the attack is successful in all cases except for the patch with the white border, and even in this case, the confidence score is significantly low at 0.45.

6 Conclusion

In this study, we addressed the often-overlooked issue of patch borders in adversarial patch research and emphasized the importance of maintaining consistent border conditions. We analyzed how patch borders affect the attack performance

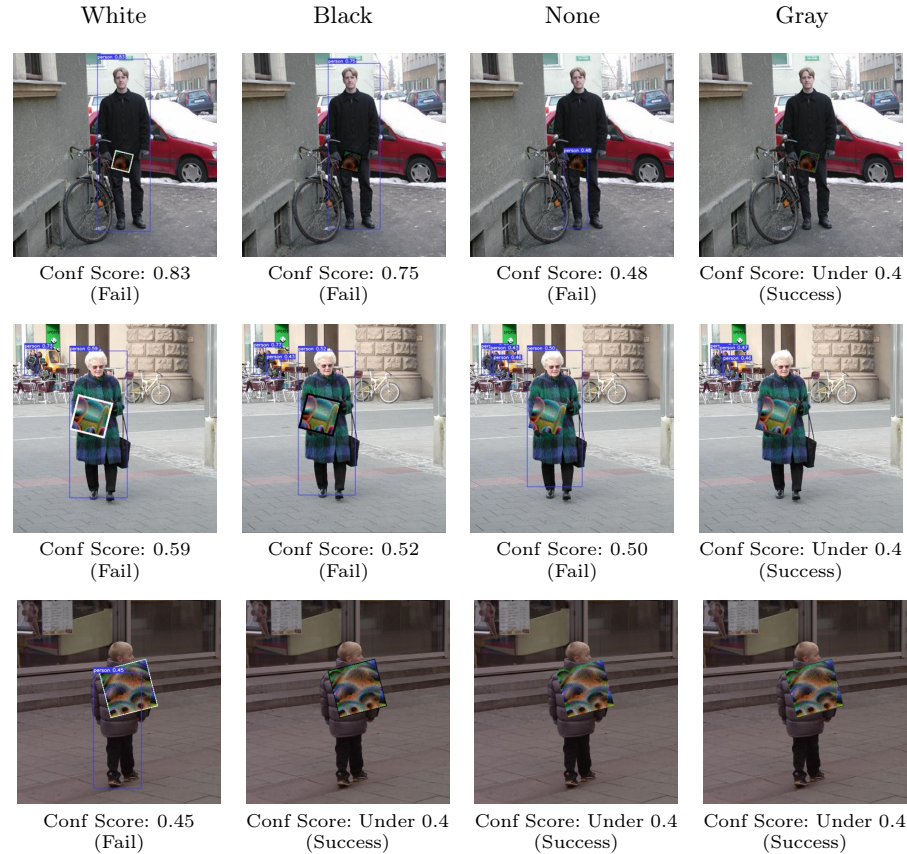


Fig. 3: Examples of hiding attack results based on border color and ratio. In each row, patches are applied with ratios of 0.1, 0.2, and 0.3 from the top to bottom rows. A lower confidence score (Conf Score) indicates a higher attack success rate. The threshold was 0.4; therefore, scores below 0.4 are considered as a “Success” and scores above 0.4 are considered as a “Failed” attack.

by performing patch attacks under different conditions. Our experiments showed that applying bicubic interpolation can increase the ASR. Moreover, patch size was not related to ASR differences, and the ASR generally increased with a border when the application ratio was between 0.1 and 0.3. Although the border thickness shows no consistent trend, we determined that using a gray border with a thickness of 4-8 pixels generally leads to an increase in ASR. Furthermore, the optimal color for maximizing ASR varies according to the dataset. Our research underscores the importance of considering characteristics in patch design and provides practical insights to enhance the effectiveness of adversarial attacks. In our study, we used a limited dataset and model for patch generation and attack. Future research should validate these strategies across a variety of datasets and different environmental conditions.

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